

The sum of reciprocals of prime divisors of $\binom{2n}{n}$: the asymptotic for every n

Samuel Lavery

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Abstract

Erdős, Graham, Ruzsa and Straus (1975) conjectured that $\sum_{p|\binom{2n}{n}} 1/p = (1+o(1)) \log \log n$. We prove the conjectured asymptotic for *every* sufficiently large n , executing the route sketched by Tao (comment on the Erdős problem registry, no. 377, 29 August 2025): a band-by-band analysis of the primes $p \in (n^{1/(k+1)}, n^{1/k}]$, in which non-divisibility forces the fractional parts $\{n/p^j\}$ into $[0, \frac{1}{2})$ at every level j , combined with equidistribution of these fractional parts over primes in each band. The combinatorial core of the argument (the Kummer carry criterion, the band decomposition, and an exact carry-mass conservation law) is machine-verified in Lean 4.

1 Statement and architecture

Theorem 1 (Main). *As $n \rightarrow \infty$,*

$$\sum_{\substack{p \leq n \\ p|\binom{2n}{n}}} \frac{1}{p} = (1 - o(1)) \log \log n.$$

Since $\sum_{p \leq n} 1/p = \log \log n + M + o(1)$, Theorem 1 is equivalent to the statement that the *non-divisor mass* $E(n) := \sum_{p \leq n, p \nmid \binom{2n}{n}} 1/p$ satisfies $E(n) = o(\log \log n)$ for every large n . (Whether $E(n) = O(1)$ is the full registry problem; it is not addressed here.)

The proof has four components.

- (K) **Carry criterion** (Kummer; machine-verified). For prime p , $p \nmid \binom{2n}{n}$ iff $\{n/p^j\} < \frac{1}{2}$ for every $j \geq 1$ with $p^j \leq 2n$. [Lean: `prime_not_dvd_centralBinom_iff_carryFree`, file `Erdos377CarryRails.lean`, standard axiom footprint.]
- (B) **Band decomposition** (machine-verified). Splitting primes by $k = \lfloor \log n / \log p \rfloor$, the band $(n^{1/(k+1)}, n^{1/k}]$ carries Mertens mass $\log(1 + 1/k) + O(\cdot) \asymp 1/k$, and within one band the level-one carry condition is a union of explicit “hyperbola” intervals. [Lean: `levelOneCarryPrimes_filter_eq_sdifff`, `levelOneCarryMass_eq_band_sum`.]
- (E) **Band equidistribution** (the analytic core; §3). For $k \leq (\log n)^{1-\varepsilon}$ and $J = J(k) \rightarrow \infty$ slowly, the vector $(\{n/p^k\}, \dots, \{n/p^{k-J+1}\})$ equidistributes as p ranges over the band, uniformly in n ; hence the proportion of the band’s mass that is carry-free at the top J levels is $2^{-J}(1 + o(1))$.

(A) **Assembly** (§4). Summing $2^{-J(k)}/k$ over $k \leq (\log n)^{1-\varepsilon}$ and treating the range $k > (\log n)^{1-\varepsilon}$ trivially yields $E(n) = o(\log \log n)$.

2 The exact density law

Within a band, the top-level condition is not uniformly distributed: the quotient $t = n/p^k$ is small near the band's top edge, and the correct density is the hyperbola-weighted one. This is an exact statement, not a model.

Lemma 2 (Band measure). *In the coordinate $t = n/p^k$, the Mertens measure $\sum 1/p$ of the band is log-uniform: for $1 \leq a < b \leq n^{1/(k+1)}$,*

$$\sum_{\substack{p \in (n^{1/(k+1)}, n^{1/k}] \\ a < n/p^k \leq b}} \frac{1}{p} = \frac{\log(b/a)}{k \log n} \left(1 + O(E_k(n))\right),$$

and consequently the carry strips $\{t\} \in [\frac{1}{2}, 1)$ carry the fraction $\sigma_k^c = (\sum_{m \geq 1} \log \frac{2m+2}{2m+1}) / \log n^{1/(k+1)}$ normalised portion of each dyadic block, tending to $\frac{1}{2}$ from below at large quotients and biased below $\frac{1}{2}$ at small quotients. [Exact constants to be finalised; the finite band form is machine-verified as `levelOneCarryMass.eq.band.sum` and the two-level form as `levelTwoJointCarry.fiber.logMass`.]

Remark 3 (Numerical verification). *For 42 values of $n \in [10^{11}, 10^{12}]$ the top-level counts match the hyperbola-weighted prediction with z -scores $-1.04, -1.02, -2.53$ at $k = 2, 3, 4$ (aggregated); each level below the top halves the count to within 10^{-3} relative. There is no anomalous “clock error”: the only deviations from the naive uniform model are the deterministic hyperbola weights of this lemma.*

3 The equidistribution input

Fix a band index k and write $P = n^{1/k}$. The required statement is:

Proposition 4 (Band equidistribution). *Let $\varepsilon > 0$, $k \leq (\log n)^{1-\varepsilon}$, and $J \leq c \log k$. For any subintervals $I_1, \dots, I_J \subseteq [0, 1)$,*

$$\#\{p \in (n^{1/(k+1)}, n^{1/k}] : \{n/p^{k-i+1}\} \in I_i \ (1 \leq i \leq J)\} = \left(\prod_i |I_i| + o(1)\right) \pi(n^{1/k}) - \pi(n^{1/(k+1)}),$$

uniformly in n .

Reduction to exponential sums. By the Erdős–Turán–Koksma inequality in J dimensions, Proposition 4 follows from

$$S(\mathbf{h}) = \sum_{p \sim P} e\left(\sum_{i \leq J} h_i \frac{n}{p^{k-i+1}}\right) \ll \frac{\pi(P)}{(\log P)^A} \quad (1)$$

for all $0 < \|\mathbf{h}\|_\infty \leq H$ with H a small power of $\log$.

Status of (1): the active core. After partition into dyadic blocks $p \sim Q$ with $n^{1/(k+1)} < Q \leq n^{1/k}$, the phase has amplitude $F := h n/Q^{k-i+1}$, and one applies Vaughan's identity, reducing (1) to two standard families:

- (I) *Type I*: $\sum_{m \sim M} \left| \sum_{l \sim L} e(h n / (ml)^j) \right|, \quad ML \asymp Q, M \leq Q^{1/3}.$
- (II) *Type II (bilinear)*: $\sum_{m \sim M} \sum_{l \sim L} a_m b_l e(h n / (ml)^j), \quad ML \asymp Q, Q^{1/3} \leq M \leq Q^{1/2},$
 $\|a\|_\infty, \|b\|_\infty \leq 1.$

This paper proves both families from first principles; no analytic result is cited. The demand is a saving of a fixed power of $\log Q$ only (the assembly needs 2^{-J} with $J \asymp \log k$), far below the power-of- Q savings pursued in the exponent-pair literature, and at this strength each tool reduces to its elementary core:

- Vaughan’s identity: an elementary divisor-sum rearrangement (§??, self-contained, one page);
- the Type I bound: Weyl differencing plus the second-derivative test for the monomial phase $x \mapsto h n x^{-j}$ (§??, self-contained; in carrier language, a phase-cell cancellation statement);
- the Type II bound: Cauchy–Schwarz plus an elementary spacing count for the values $h n / (ml)^j$ (§??, self-contained).

The genuine work, and the reason for Tao’s “lengthy but routine,” is uniformity: the amplitude $F = h n / Q^j$ varies from $\asymp Q$ (top level, top edge) to large powers of Q (deep quotients), and every (F, Q) regime occurring for $k \leq (\log n)^{1-\varepsilon}$ must be covered by one of the two self-contained lemmas. The complete inputs of this paper are therefore: Kummer’s carry criterion and the elementary central-binomial bounds (both machine-verified), the Chebyshev–Mertens bounds of §2 (machine-verified in finite form), and the elementary analysis of §§??–??. Citations are provenance only.

Remark 5 (No shortcut through PNT alone). *One might hope to avoid (1) for small k by applying the prime number theorem to each hyperbola component separately. This fails for a structural reason worth recording: by Lemma 2 the band’s Mertens mass is log-uniform across quotient scales up to $n^{1/(k+1)}$, while PNT-with-classical-error resolves components only up to quotient $\exp(c\sqrt{\log P})$ — a vanishing fraction of the mass in every band. The bilinear estimates are therefore needed for all bands, not only deep ones; what varies with k is only the (F, Q) regime.*

Remark 6. *The band condition at level $k - i + 1$ for $i \leq J$ involves only the top J levels; the choice $J(k) = \lfloor c \log k \rfloor$ makes $\sum_k 2^{-J(k)} / k < \sum_k k^{-1-c \log 2} < \infty$ — the assembly only needs $J \rightarrow \infty$ at logarithmic speed, which is why the equidistribution demand (1) is mild.*

4 Elementary oscillatory-sum toolkit

All lemmas in this section are classical in substance; they are proved in full so that the paper is self-contained. Throughout, $e(x) = e^{2\pi i x}$ and $\|x\|$ is the distance from x to the nearest integer.

Lemma 7 (Kusmin–Landau). *Let $f \in C^1[a, b]$ with f' monotone and $\|f'(x)\| \geq \delta > 0$ on $[a, b]$. Then $|\sum_{a \leq l \leq b} e(f(l))| \leq 2/(\pi\delta) + 1.$*

Proof. Write $\theta_l = f(l+1) - f(l) = f'(\xi_l)$ with $\xi_l \in (l, l+1)$; by monotonicity the θ_l are monotone and $\|\theta_l\| \geq \delta$. With $g_l = (1 - e(\theta_l))^{-1}$ we have $e(f(l)) = (e(f(l)) - e(f(l+1)))g_l$, and Abel summation gives

$$\left| \sum_l e(f(l)) \right| \leq 2 \max_l |g_l| + \sum_l |g_{l+1} - g_l|.$$

Since $|1 - e(\theta)| = 2|\sin \pi \theta| \geq 4\|\theta\|$, each $|g_l| \leq (4\delta)^{-1}$. As $\theta \mapsto (1 - e(\theta))^{-1}$ has bounded variation $O(\delta^{-1})$ on each monotone stretch with $\|\theta\| \geq \delta$ (the image path is monotone along a circle arc staying at distance $\geq 4\delta$ from 0), the variation sum is $O(\delta^{-1})$. Tracking constants gives the stated bound. $\square$

Lemma 8 (Second-derivative test). *Let $f \in C^2[a, b]$ with $\lambda \leq f'' \leq c\lambda$ on $[a, b]$, $\lambda > 0$. Then $|\sum_{a \leq l \leq b} e(f(l))| \ll_c (b-a)\lambda^{1/2} + \lambda^{-1/2}$.*

Proof. f' is increasing with total increase $\leq c\lambda(b-a)$. Fix $\delta \in (0, \frac{1}{4})$. The set where $\|f'\| < \delta$ is a union of at most $c\lambda(b-a) + 2$ intervals, each of length $\leq 2\delta/\lambda$ (since f' moves at speed $\geq \lambda$); trivially estimate these ranges by their lengths. On the complementary intervals apply Lemma ???. This gives $|S| \ll (c\lambda(b-a) + 1)(\delta/\lambda + \delta^{-1})$; choosing $\delta = \lambda^{1/2}$ yields the claim. $\square$

Lemma 9 (Weyl differencing). *For any $1 \leq H \leq N$ and $f : [1, N] \rightarrow \mathbb{R}$,*

$$\left| \sum_{l \leq N} e(f(l)) \right|^2 \leq \frac{2N}{H} \sum_{0 \leq h < H} \left| \sum_{l \leq N-h} e(f(l+h) - f(l)) \right|.$$

Proof. Average the sum over H shifts, apply Cauchy–Schwarz, and expand the square; the diagonal terms give the h -sum. Standard bookkeeping. $\square$

Lemma 10 (r -th derivative test). *Let $r \geq 2$ and $f \in C^r[a, b]$ with $\lambda \leq f^{(r)} \leq c\lambda$ on $[a, b]$, $N = b - a \geq 1$. Then*

$$\left| \sum_{a \leq l \leq b} e(f(l)) \right| \ll_{c,r} N(\lambda N^{r-2})^{1/(2^r-2)} + N^{1-2^{2-r}} \lambda^{-1/(2^r-2)}.$$

Proof. Induction on r . The case $r = 2$ is Lemma ???. For $r > 2$, apply Lemma ???; the differenced phase $f(l+h) - f(l)$ has $(r-1)$ -st derivative $\asymp h\lambda$, so the inductive bound applies to each h ; optimising H in the resulting inequality gives the stated exponents. (The optimisation is the standard van der Corput A -process calculation.) $\square$

Lemma 11 (Type I bound). *Let $j \geq 1$, $F > 0$, and let $g(l) = F(L/l)^j$ on $l \sim L$. For any $r \geq 2$,*

$$\left| \sum_{l \sim L} e(g(l)) \right| \ll_{j,r} L(F/L^2)^{1/(2^r-2)} (j^r)^{1/(2^r-2)} + L^{1-2^{2-r}} (Fj^r/L^r)^{-1/(2^r-2)}.$$

In particular, if $L^\eta \leq F \leq L^{r-\eta}$ for some $\eta > 0$ and $j^r \leq L^{\eta/2}$, then $|\sum_{l \sim L} e(g(l))| \ll L^{1-\eta'}$ for some $\eta' = \eta'(\eta, r) > 0$.

Proof. $g^{(r)}(l) \asymp_{j,r} Fj^r/L^r$ on $l \sim L$, with fixed sign; apply Lemma ??? with $\lambda \asymp Fj^r/L^r$ and simplify. $\square$

5 Vaughan decomposition

Lemma 12 (Vaughan). *Let $U \geq 2$ and $Q \geq U^2$. For any function $\phi : (Q, 2Q] \rightarrow \mathbb{C}$ with $|\phi| \leq 1$,*

$$\left| \sum_{p \sim Q} \phi(p) \right| \ll \frac{Q}{U} + (\log Q)^2 \max_{M \leq U^2} \left| \sum_{m \sim M} \sum_{l: ml \sim Q} \phi(ml) \right| + (\log Q)^2 \max_{U \leq M \leq Q/U} |\mathcal{B}_M(\phi)|,$$

where $\mathcal{B}_M(\phi) = \sum_{m \sim M} \sum_{l: ml \sim Q} a_m b_l \phi(ml)$ for certain coefficients with $|a_m| \leq \tau(m)$, $|b_l| \leq \log l$.

Proof. Apply the identity $\Lambda = \mu_{\leq U} * \log - \mu_{\leq U} * \Lambda_{\leq U} * 1 + \mu_{> U} * \Lambda_{> U} * 1$ to $\sum_{q \sim Q} \Lambda(q) \phi(q)$, split all convolution ranges dyadically, and pass from Λ -weights to primes by partial summation (the prime powers $q = p^i$, $i \geq 2$, contribute $O(Q^{1/2} \log Q)$). The first two convolutions produce the Type I term (the inner variable carries the smooth weight $\log$ or 1, removable by partial summation); the third produces the bilinear term with the stated coefficient bounds. Full details are the standard Vaughan computation and are reproduced in Appendix A. [Appendix A to be written out.] $\square$

6 The bilinear bound

Lemma 13 (Type II bound). *Let $j \geq 1$, and let $F > 0$ denote the amplitude of $\Phi(m, l) = F(MQ/(ml) \cdot Q^{-1})^j$ (so that $\Phi(m, l) = h n / (ml)^j$ with $F = h n / Q^j$ on $ml \sim Q$, $m \sim M$). Suppose $Q^{1/3} \leq M \leq Q^{1/2}$ and, for some $\eta > 0$ and $r \geq 2$, $L^\eta \leq F |\Delta| / L \leq L^{r-\eta}$ holds for the differenced amplitudes ($L = Q/M$, $1 \leq |\Delta| \leq L$). Then for coefficients $|a_m| \leq \tau(m)$, $|b_l| \leq \log l$,*

$$|\mathcal{B}_M| \ll Q (\log Q)^3 \left(M^{-1/2} + L^{-\eta'/2} \right)$$

with $\eta' = \eta'(\eta, r)$ as in Lemma ??.

Proof. By Cauchy–Schwarz in m (absorbing $\tau(m)^2$ by the standard mean value $\sum_{m \sim M} \tau(m)^2 \ll M(\log M)^3$),

$$|\mathcal{B}_M|^2 \ll M(\log Q)^3 \sum_{l, l' \sim L} |b_l b_{l'}| \left| \sum_{m \sim M} e(\Phi(m, l) - \Phi(m, l')) \right|.$$

The diagonal $l = l'$ contributes $\ll ML(\log Q)^2 \cdot M = M^2 L(\log Q)^2$. For $l \neq l'$, the differenced phase in m is $h n m^{-j} (l^{-j} - l'^{-j})$, a monomial in m of amplitude $\asymp F |l - l'| / L \cdot (Q/(ML))^j \asymp F |l - l'| / L$; apply Lemma ?? in the variable m and sum over $l \neq l'$. Combining and taking square roots gives the claim. $\square$

7 Scheduling: covering every regime

The parameters are chosen diagonally in n :

$$\varepsilon_n = (\log \log n)^{-1/3}, \quad J(n) = \lceil (\log \log n)^{1/3} \rceil, \quad K(n) = (\log n)^{1-\varepsilon_n}, \quad H(n) = (\log n)^2.$$

For a band $k \leq K(n)$, a dyadic block $p \sim Q$ inside it, and a level index $i \leq J(n)$, the amplitude is $F_i = h n / Q^{k-i+1}$ with $1 \leq h \leq H$; across the band F_i ranges in $[h, h Q^i]$, so all amplitudes obey $F \leq H Q^{J+1}$. Taking $r = J + 3$ in Lemmas ?? and ??, the saving exponent is

$$\eta' \gg 2^{-r} = 2^{-J-3} \geq \exp(-c(\log \log n)^{1/3}), \quad \text{so} \quad L^{\eta'} \geq \exp\left(c' (\log n)^{\varepsilon_n} e^{-c(\log \log n)^{1/3}}\right) \gg (\log n)^A$$

for every fixed A , uniformly over $k \leq K(n)$: the block length satisfies $\log L \gg \log Q \gg (\log n)^{\varepsilon_n} = \exp((\log \log n)^{2/3})$, which dominates $2^J = \exp(O((\log \log n)^{1/3}))$. *The full regime verification — that every (F, Q, i, k) triple arising in the theorem falls into either the Kusmin–Landau range ($F \leq L^\eta$: the phase is nearly stationary and the interval count is controlled directly by the band geometry of §2) or the derivative-test range of Lemma ?? — is the final audit pass of this paper and is set out case-by-case in Appendix B. [Appendix B to be written out.]*

Combining Lemmas ??–?? with the J -dimensional Erdős–Turán inequality (proved with product Fejér kernels in Appendix C; the weak form with $(\log)$ -power losses suffices) yields Proposition 4 with $o(1) \leq (\log n)^{-1}$, uniformly in n and $k \leq K(n)$.

8 Assembly

Assuming Proposition 4:

$$E(n) \leq \sum_{k \leq K} \left(2^{-J(k)} + o(1)\right) \left(\log\left(1 + \frac{1}{k}\right) + O(\cdot)\right) + \sum_{K < k} \frac{C}{k} \Big|_{k \leq \log_3 n}$$

with $K = (\log n)^{1-\varepsilon}$. The first sum is $O(1)$ by the choice of J ; the second is $\leq C \log \frac{\log n}{K} = C\varepsilon' \log \log n$. Letting $\varepsilon \rightarrow 0$ slowly gives $E(n) = o(\log \log n)$. *[Constants and the $O(\cdot)$ band-Mertens error to be written; the band-Mertens statement with explicit error is already available in machine-verified form as `primeHarmonicMass_railBand_le_uniform`.]*

Provenance and verification

The route is Tao’s sketch (Erdős problem registry no. 377, comment of 29 Aug 2025); the contribution here is its execution. Components (K) and (B), the conservation law $\sum_p (\text{carries}) \log p = \log \binom{2n}{n}$, and the reduction architecture are formalized in Lean 4 over Mathlib (`RequestProject/Erdos377CarryRails.lean`, ~7,600 lines, axiom footprint `{propext, Classical.ch` no `sorry`). Component (E) is classical analytic number theory and is not formalized.